B.1 DIMETHYL ETHER (DME) PRODUCTION, UNIT 200

DME is used primarily as an aerosol propellant. It is miscible with most organic solvents, has a high solubility in water, and is completely miscible in water and 6% ethanol [1]. Recently, the use of DME as a fuel additive for diesel engines has been investigated due to its high volatility (desirable for cold starting) and high cetane number. The production of DME is via the catalytic dehydration of methanol over an acid zeolite catalyst. The main reaction is

$$\underset{methanol}{2CH_3OH} \rightarrow \underset{DME}{(CH_3)_2O} + H_2O \tag{B.1.1}$$

In the temperature range of normal operation, there are no significant side reactions.

B.1.1 Process Description

A preliminary process flow diagram for a DME process is shown in Figure B.1.1, in which 50,000 metric tons per year of 99.5 wt% purity DME product is produced. Due to the simplicity of the process, an operating factor greater than 0.95 (8375 h/y) is used.

Fresh methanol, Stream 1, is combined with recycled reactant, Stream 13, and vaporized prior to being sent to a fixed-bed reactor operating between 250°C and 370°C. The single-pass conversion of methanol in the reactor is 80%. The reactor effluent, Stream 7, is then cooled prior to being sent to the first of two distillation columns: T-201 and T-202. DME product is taken overhead from the first column. The second column separates the water from the unused methanol. The methanol is recycled back to the front end of the process, and the water is sent to wastewater treatment to remove trace amounts of organic compounds.

Stream summaries, utility summaries, and equipment summaries are presented in Tables B.1.1–B.1.3.

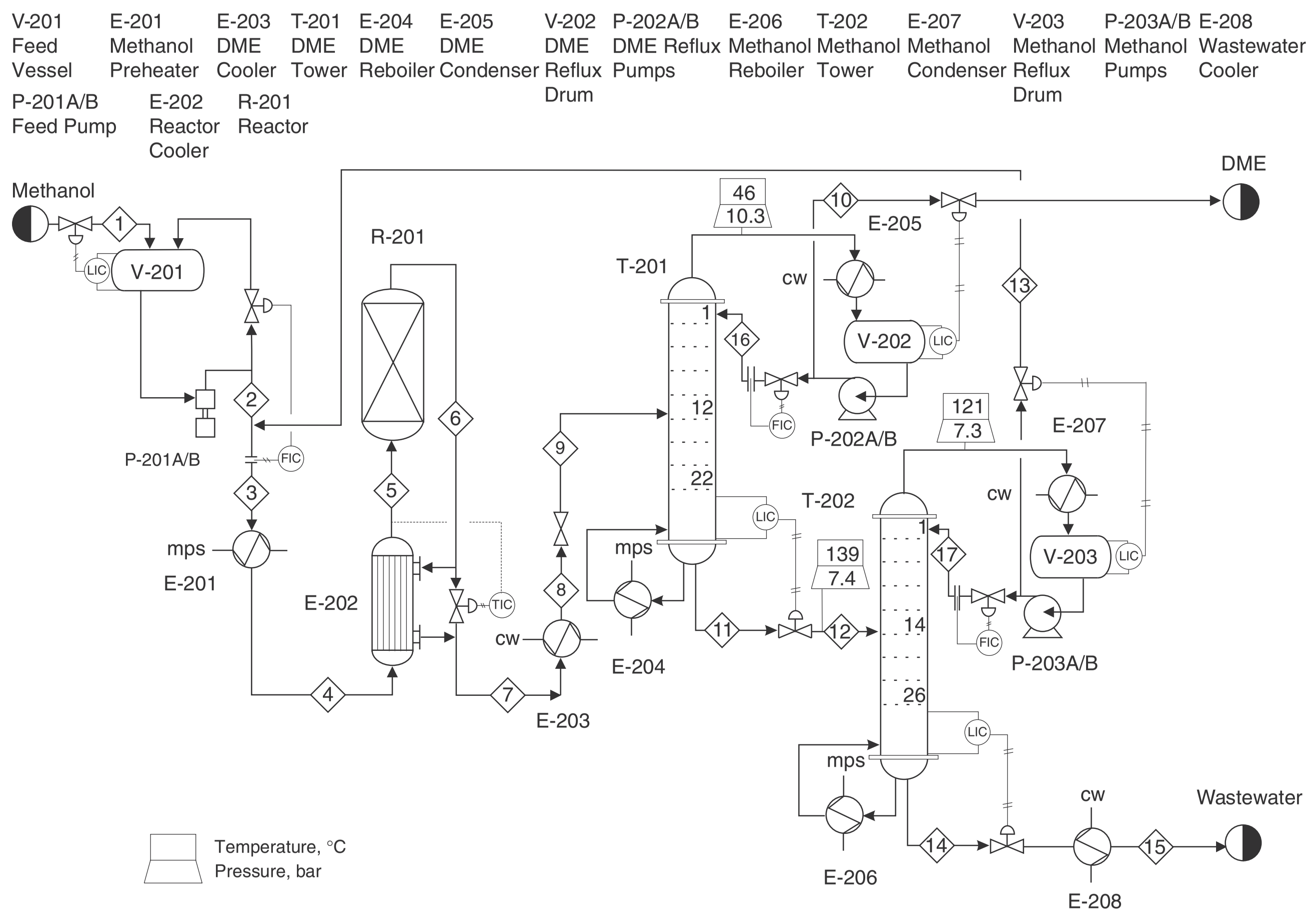

Figure B.1.1 Unit 200: Dimethyl Ether Process Flow Diagram

Table B.1.1 Stream Table for Unit 200

Stream Number	1	2	3	4	5	6	7	8	
Temperature (°C)	25	25	45	154	250	365	278	100	
Pressure (bar)	1.0	15.5	15.2	15.1	14.7	13.9	13.8	13.4	
Vapor fraction	0.0	0.0	0.0	1.0	1.0	1.0	1.0	0.08	
Mass flow (tonne/h)	8.37	8.37	10.49	10.49	10.49	10.49	10.49	10.49	
Mole flow (kmol/h)	262.2	262.2	328.3	328.3	328.3	328.3	328.3	328.3	
Component flowrates (kmol/h)									
Dimethyl ether	0.0	0.0	1.44	1.44	1.44	130.63	130.63	130.63	
Methanol	259.7	259.7	323.24	323.24	323.24	64.85	64.85	64.85	
Water	2.5	2.5	3.80	3.80	3.80	133.00	133.00	133.00	
Stream Number	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
Temperature (°C)	89	45.8	151.8	138	118	166	50	46	118
Pressure (bar)	10.4	11.4	11.2	7.4	15.5	7.4	1.2	10.3	7.3
Vapor fraction	0.252	0.0	0.0	0.04	0.0	0.0	0.0	0.0	0.0
Mass flow (tonne/h)	10.49	5.98	4.52	4.51	2.12	2.39	2.39	3.32	9.71
Mole flow (kmol/h)	328.3	130.0	198.5	198.5	66.1	132.4	132.4	72.2	302.6
Component flowrates (kmol/h)									
Dimethyl ether	130.63	129.20	1.44	1.44	1.44	0.00	0.00	71.9	6.4
Methanol	64.85	0.60	64.25	64.25	63.54	0.71	0.71	0.3	290.2
Water	133.00	0.01	132.99	132.99	1.30	131.69	131.69	0.0	6.0

Table B.1.2 Utility Summary Table for Unit 200

E-201	E-203	E-204	E-205	E-206	E-207	E-208
mps	cw	mps	cw	mps	cw	cw
7192 kg/h	286,800 kg/h	1256 kg/h	83,940 kg/h	6130 kg/h	294,250 kg/h	28,250 kg/h

Table B.1.3 Major Equipment Summary for Unit 200

Heat Exchangers	
E-201 A = 99.4 m^2 Floating head, carbon steel, shell-and-tube design Process stream in tubes Q = 14,400 MJ/h Maximum pressure rating of 15 bar	**E-205** A = 100.6 m^2 Fixed head, carbon steel, shell-and-tube design Process stream in shell Q = 3580 MJ/h Maximum pressure rating of 10 bar
E-202 A = 171.0 m^2 Floating head, carbon steel, shell-and-tube design Process stream in tubes and shell Q = 2030 MJ/h Maximum pressure rating of 15 bar	**E-206** A = 83.0 m^2 Floating head, carbon steel, shell-and-tube design Process stream in shell Q = 12,180 MJ/h Maximum pressure rating of 11 bar
E-203 A = 101.8 m^2 Floating head, carbon steel, shell-and-tube design Process stream in shell Q = 11,980 MJ/h Maximum pressure rating of 14 bar	**E-207** A = 22.7m^2 Floating head, carbon steel, shell-and-tube design Process stream in shell Q = 12,360 MJ/h Maximum pressure rating of 7 bar
E-204 A = 22.0 m^2 Floating head, carbon steel, shell-and-tube design Process stream in shell Q = 2490 MJ/h Maximum pressure rating of 11 bar	**E-208** A = 22.8 m^2 Floating head, carbon steel, shell-and-tube design Process stream in shell Q = 1180 MJ/h Maximum pressure rating of 8 bar
Pumps	
P-201 A/B Reciprocating/electric drive Carbon steel Power = 7.2 kW (actual) 60% efficient Pressure out = 15.5 bar	**P-202 A/B** Centrifugal/electric drive Carbon steel Power = 5.2 kW (actual) 40% efficient Pressure out = 16 bar
P-202 A/B Centrifugal/electric drive Carbon steel Power = 1.0 kW (actual) 40% efficient Pressure out = 11.4 bar	

(continued)

Table B.1.3 Major Equipment Summary for Unit 200 (*Continued*)

Reactor **R-201** Carbon steel Packed-bed section 13.3 m high filled with catalyst Diameter = 1.33 m Height = 15 m Maximum pressure rating of 14.7 bar	
Towers **T-201** Carbon steel 22 SS valve trays plus reboiler and condenser 24-in tray spacing Column height = 15.8 m Diameter = 0.79 m Maximum pressure rating of 10.6 bar	**T-202** Carbon steel 26 SS valve trays plus reboiler and condenser 18-in tray spacing Column height = 14.9 m Diameter = 0.87 m Maximum pressure rating of 7.3 bar
Vessels **V-201** Horizontal Carbon steel Length = 3.42 m Diameter = 1.14 m Maximum pressure rating of 1.1 bar **V-202** Horizontal Carbon steel Length = 2.89 m Diameter = 0.98 m Maximum pressure rating of 10.3 bar	**V-203** Horizontal Carbon steel Length = 2.53 m Diameter = 0.85 m Maximum pressure rating of 7.3 bar

B.1.2 Reaction Kinetics

The reaction taking place is mildly exothermic with a standard heat of reaction, $\Delta H_{reac}(25°C) = -11{,}770$ kJ/kmol. The equilibrium constant for this reaction at three different temperatures is given in Table B.1.4.

Table B.1.4 Equilibrium Constant for Methanol Dehydration Reaction

T	*K*
473 K (200°C)	33.8
573 K (300°C)	12.4
673 K (400°C)	6.17

Using the data in Table B.1.4, the equilibrium constant (K) can be fitted to the following equation, $K = 0.1103\exp\left[\frac{2708.6}{T[K]}\right]$. The corresponding equilibrium conversions for pure methanol feed over the above temperature range are greater than 80%.